BAHIR DAR UNIVERSITY

Bahir Dar Institute of Technology

Faculty of Computing

OSSP PROJECT

Operating System and System Programming

Ubuntu Server 24.04.4 LTS — Individual Assignment

Department	Software Engineering
Submitted By	Meseret Yilak
ID number	BDU1604608
OS Assigned	Ubuntu Server 24.04.4 LTS
Assignment Type	Individual
Submitted to	Lec .Wendimu
Year	2nd Year, 2018 E.C
Section	A

Submission Date: 20/08/2018 E.C

TABLE OF CONTENTS

1. Introduction ..... ….3
   1.1 Background 3
   1.2 Motivation ..... ….3
2. Objectives 4
3. Requirements .....…….. 4
   3.1 Hardware Requirements .....…..... 4
   3.2 Software Requirements .....… .. 5
4. Installation Steps .....……... 5
   4.1 Download Ubuntu Server ISO .....… .. .5
   4.2 Install VirtualBox ..... ……..5
   4.3 Create Virtual Machine ..... …….6
   4.4 Attach ISO File .....…….. 7
   4.5 Boot Process .....…….. 7
   4.6 Language & Keyboard .....……..... 8
   4.7 Network Configuration ..... ……8
   4.8 Disk Partitioning ..... ……9
   4.9 User Account Setup ..... ……..9
   4.10 SSH Configuration ..... ……10
   4.11 Installation Process ..... …….11
   4.12 Reboot & Login ..... ……..11
5. Issues Encountered ..... ……..12
6. Solutions ..... ……13
7. Filesystem Support .....…..... ……..14
8. Advantages and Disadvantages ....…..... ……14
9. Conclusion ..... ……15
10. Future Recommendations .....……..... ……16
11. Virtualization in Modern OS .....…….. ……..17
   11.1 What is Virtualization .....…….. ……17
   11.2 Why Virtualization ..... ……..17
   11.3 How Virtualization Works ..... …….. 19
12. References ..... ……..21

Ubuntu Server OS Installation in Virtual Environment

a. Introduction

Background

Operating systems are the backbone of every computing environment. As future software engineers, it becomes critical to understand not just how to use an operating system, but how to install, configure, and manage one from the ground up. Virtual environments offer a safe, controlled, and cost-effective way to practice this — without putting any physical hardware at risk.

Ubuntu Server is a Debian-based Linux distribution maintained by Canonical Ltd. First released in 2004, Ubuntu has grown into one of the most widely deployed server operating systems in the world. The Long-Term Support (LTS) editions, released every two years, are especially popular in enterprise and academic environments because they guarantee five years of security patches and updates. Ubuntu Server 24.04.4 LTS (codenamed 'Noble Numbat') was released in 2024 and represents the most current stable LTS release at the time of this project.

Motivation

The motivation behind choosing Ubuntu Server as the subject of this assignment is rooted in several practical considerations. First, Ubuntu Server is completely free and open-source, making it accessible to students and institutions without licensing costs. Second, it has an enormous community, which means troubleshooting resources, tutorials, and documentation are abundantly available. Third, most cloud platforms — including AWS, Google Cloud, and Azure — offer Ubuntu Server as a first-class operating system for virtual machines, meaning the skills learned here directly translate to real-world cloud deployments. Finally, working in a headless server environment (no graphical desktop) forces the user to develop genuine command-line proficiency, which is a core skill for any systems or software engineer.

b. Objectives

The following objectives guided this project:

1. To successfully install Ubuntu Server 24.04.4 LTS inside Oracle VM VirtualBox on a host Windows machine.
2. To document each step of the installation process with screenshots and clear explanations.
3. To understand the concept of virtual environments and how they isolate guest operating systems from the host.
4. To explore the filesystem structure and understand which filesystem types are supported by Ubuntu Server and why ext4 is the default choice.
5. To evaluate the advantages and disadvantages of Ubuntu Server compared to alternatives.
6. To develop foundational skills in Linux system administration that are essential for a software engineering career.

c. Requirements

i. Hardware Requirements

The following hardware specifications were used for this installation. These represent the minimum recommended requirements for running Ubuntu Server inside VirtualBox comfortably:

Component	Minimum Requirement	Used in This Setup
Processor (CPU)	Dual-core, 64-bit processor	Intel Core i5 (64-bit, VT-x enabled)
RAM (Host Machine)	4 GB minimum	8 GB host RAM
RAM (Allocated to VM)	1 GB minimum	2 GB allocated
Disk Space (Host)	20 GB free	50 GB free on host
Disk Space (VM)	10 GB minimum	20 GB virtual disk
Architecture	x86_64 (64-bit)	x86_64
Virtualization Support	VT-x or AMD-V in BIOS	VT-x enabled in BIOS
Network Interface	One NIC (NAT or Bridged)	NAT (enp0s3)

ii. Software Requirements

Software	Version / Details	Purpose
Host OS	Windows 10/11 (64-bit)	The operating system running on the physical machine
Oracle VM VirtualBox	Version 7.x	Hypervisor / Virtual machine manager
VirtualBox Extension Pack	Matching VirtualBox version	Enables USB 2.0/3.0, RDP, disk encryption
Ubuntu Server ISO	ubuntu-24.04.4-live-server-amd64.iso	Installation image for the guest OS

d. Installation Steps

The following steps were carried out to install Ubuntu Server 24.04.4 LTS inside Oracle VM VirtualBox. Screenshots have been taken at each key stage and can be inserted into the spaces provided below.

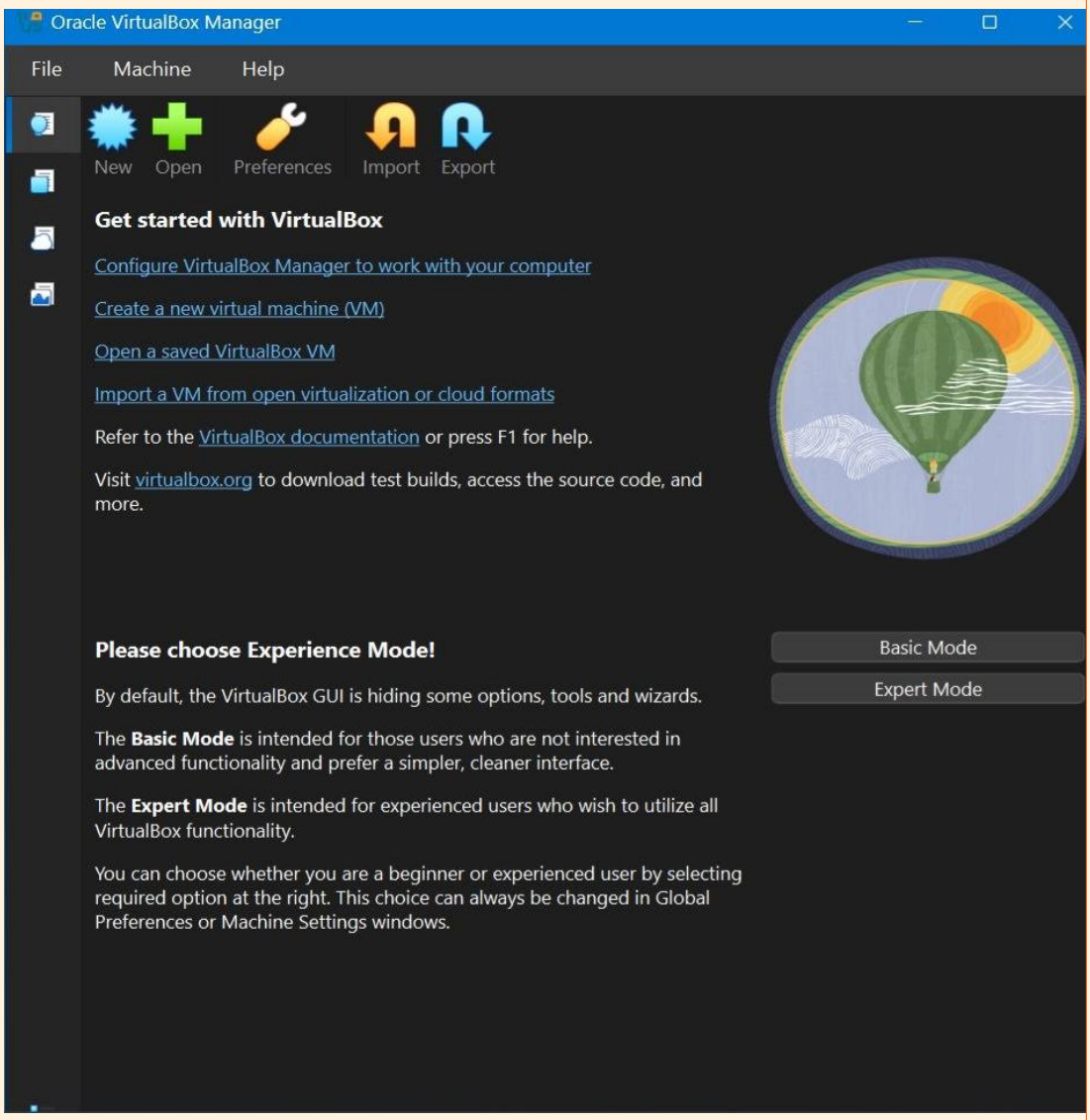
Step 1 .Download the Ubuntu Server ISO

Visit the official Ubuntu website at https://ubuntu.com/download/server and download the Ubuntu Server 24.04.4 LTS ISO image (ubuntu-24.04.4-live-server-amd64.iso). The file size is approximately 3 GB. Verify the checksum after download to ensure file integrity.

Step 2 . Install and Open Oracle VM VirtualBox

Download Oracle VM VirtualBox from https://www.virtualbox.org and install it on the host machine. After installation, launch VirtualBox. The main window shows a list of existing virtual machines (if any) on the left, and action buttons on the right.

VirtualBox Main Window



Step 3 .Create a New Virtual Machine

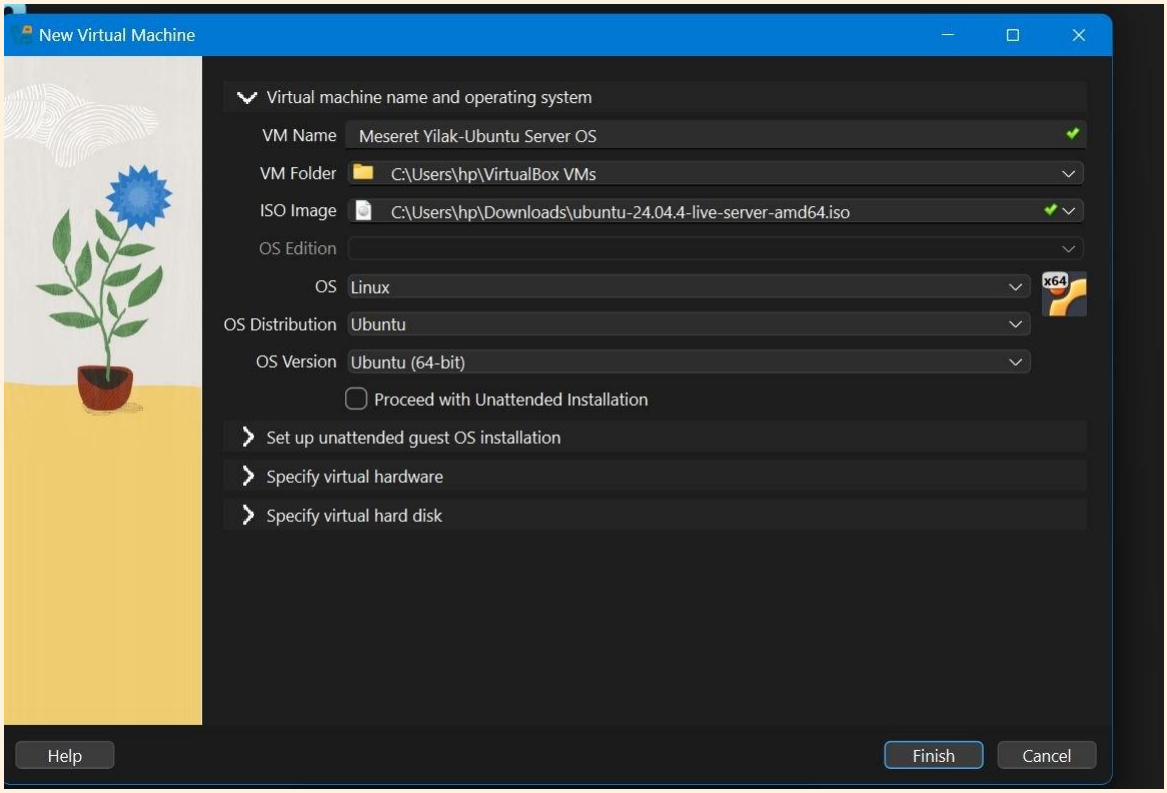
Click the 'New' button to open the VM creation wizard. Fill in the details as follows:

- Name: ubuntu-server-os (the name visible in your VirtualBox list)
- Type: Linux
- Version: Ubuntu (64-bit)
- Memory: 2048 MB (2 GB)
- Hard Disk: Create a virtual hard disk now, 20 GB, VDI format, dynamically allocated

Step 4 .Attach the Ubuntu Server ISO

Before booting the VM for the first time, attach the downloaded ISO file as the boot media. Go to the VM Settings → Storage → under 'Controller: IDE', click the empty disc icon → click the disc icon on the right and select 'Choose a disk file' → navigate to and select your downloaded ubuntu-24.04.4-live-server-amd64.iso file.

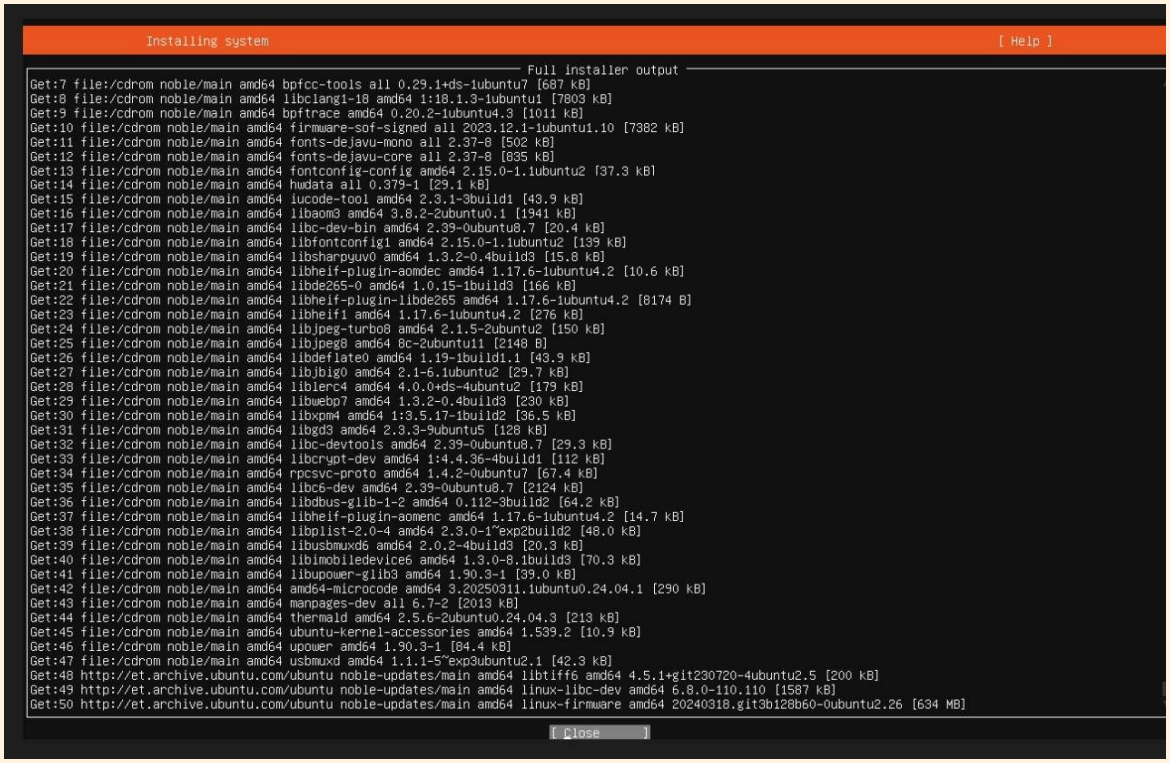
ISO Attachment



Step 5 .Boot the Virtual Machine

Double-click the VM in VirtualBox to start it. The VM will boot from the attached ISO. After a moment, the GRUB boot menu appears. Select 'Try or Install Ubuntu Server' and press Enter. The installation kernel loads and brings up the Ubuntu Server installer (called 'Subiquity').

Boot Menu / Installer Loading



Step 6 . Language and Keyboard Selection

The Subiquity installer first asks you to select a language. Choose 'English' for consistency with most documentation. Next, confirm or set the keyboard layout to match your physical keyboard (e.g., English US). Press Enter/Done to continue.

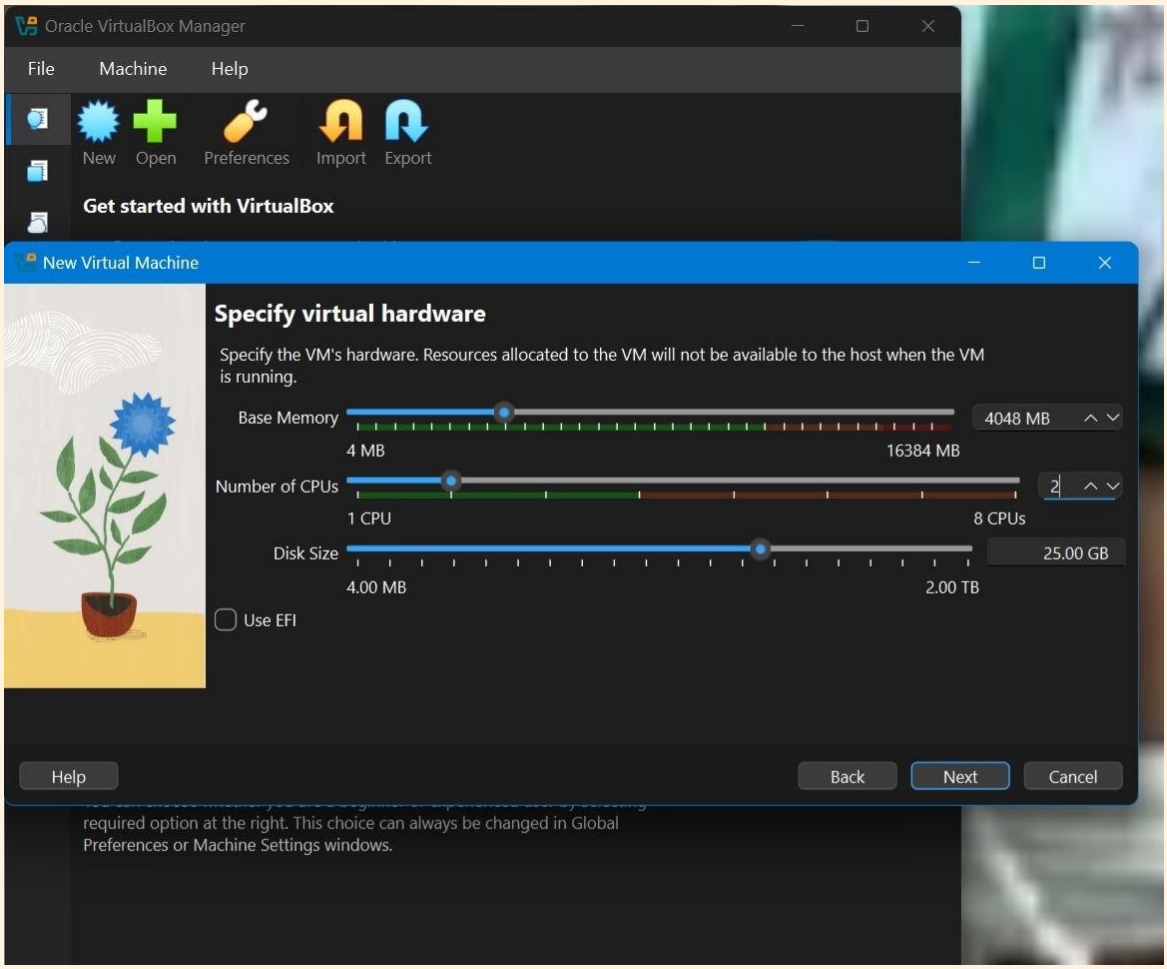
Step 7 . Network Configuration

The installer detects the virtual network interface (enp0s3, using NAT in VirtualBox). Leave the default DHCP settings — the VM will automatically receive an IP address from VirtualBox's internal DHCP server. The IPv4 address assigned is 10.0.2.15, which is VirtualBox's standard NAT address. Press Done.

Step 8 . Disk Partitioning

Choose 'Use an entire disk' for simplicity. The installer selects the virtual 25 GB disk (VBOX_HARDDISK) automatically. It proposes a layout with a small EFI partition, a main ext4 root partition (/), and a swap area. Accept the default layout and confirm when prompted that this will erase the virtual disk. Since this is a VM, there is nothing to lose — confirm and continue.

Disk Partitioning



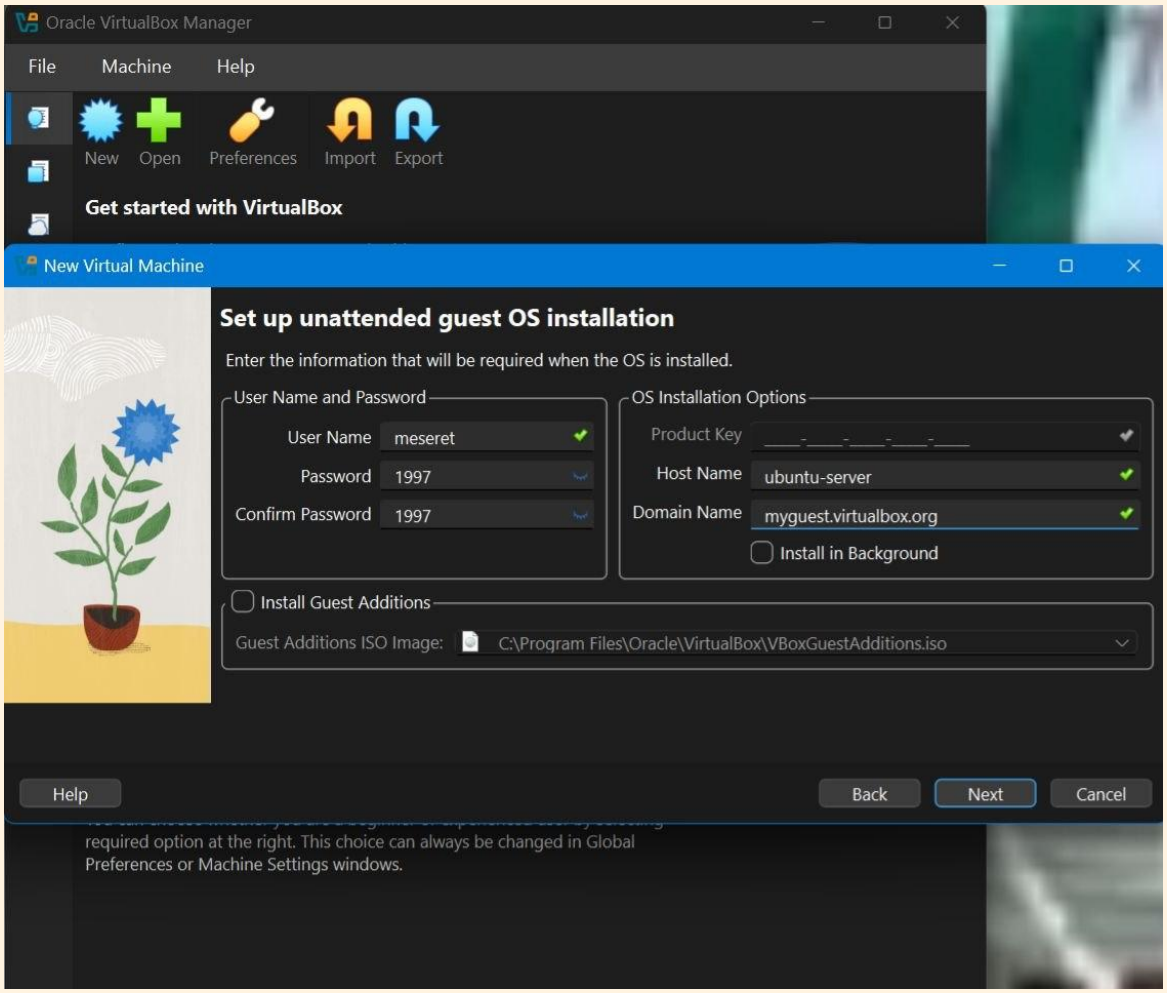
Step 9 . User Account Creation

This is a critical step per the assignment instructions — the user account must be created with your full name. Fill in the following fields:

- Your name: [Your Full Name — e.g., Meseret Yilak]
- Server name: ubuntu-server-os

- Username: meseret (lowercase, no spaces)
- Password: [Create a strong password]
- Confirm password: [Re-enter the same password]

User Account Creation



Step 10 . SSH Server Setup

The installer offers to install OpenSSH Server. For a server environment, this is extremely useful as it allows remote terminal access (e.g., from Windows using PuTTY or Windows Terminal). Check the box to 'Install OpenSSH server' and press Done. Skip any additional snap packages for now.

Step 11 . Installation Begins

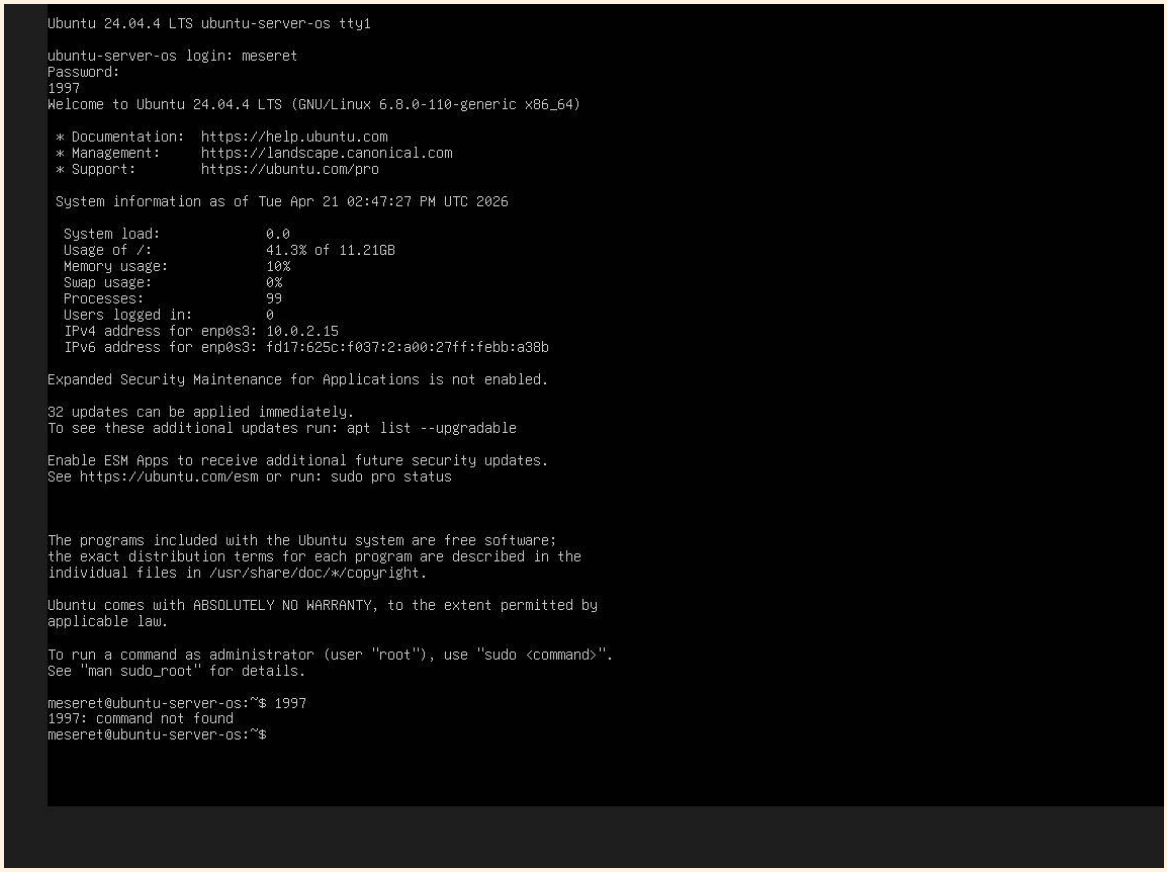
The installer now proceeds to install the base system, kernel, and selected packages. This process typically takes 20–40 minutes depending on the host machine's speed. A progress bar shows the current installation stage (downloading updates, installing packages, installing kernel, etc.).

Step 12 . Reboot and First Login

Once installation completes, the installer prompts 'Please remove the installation medium and press ENTER'. In VirtualBox, go to Devices → Optical Drives and unmount the ISO, then press Enter. The VM reboots. After rebooting, the terminal login prompt appears. Log in with the username and password created in Step 9.

After successful login, a welcome message is displayed showing system information — exactly as captured in the photo submitted with this assignment:

Successful Login (Provided Photo)



```
Ubuntu 24.04.4 LTS ubuntu-server-os tty1

ubuntu-server-os login: meseret
Password:
1997
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Apr 21 02:47:27 PM UTC 2026

  System load:             0.0
  Usage of /:              41.3% of 11.21GB
  Memory usage:            10%
  Swap usage:              0%
  Processes:               99
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:febb:a38b

Expanded Security Maintenance for Applications is not enabled.

32 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

meseret@ubuntu-server-os:~$ 1997
1997: command not found
meseret@ubuntu-server-os:~$
```

The system information visible after login confirms the installation was successful:

System Parameter	Value	Meaning
OS Version	Ubuntu 24.04.4 LTS	Long-term support release, supported until 2029
Kernel	GNU/Linux 6.8.0-110-generic x86_64	64-bit Linux kernel version 6.8
System Load	0.32	Low CPU load — system is idle and healthy
Disk Usage (/)	41.5% of 11.21 GB	About 4.6 GB used out of the allocated virtual disk
Memory Usage	11%	Low RAM consumption — server OS has no GUI overhead
Swap Usage	0%	No swap being used — memory is sufficient
Processes	109	Active processes running on the system
IPv4 Address	10.0.2.15 (enp0s3)	NAT address assigned by VirtualBox DHCP

e. Issues (Problems Faced)

During the installation process, the following issues were encountered:

Issue 1 .Slow Installation Due to Mirror Updates

During the package installation phase, the installer attempted to download updated package lists from Ubuntu's online mirrors. Because the VM was connected to the internet through NAT, this worked but was slow, adding several minutes to the installation time.

Issue 2 .Screen Resolution Limitation

After installation, the VirtualBox window showed the Ubuntu Server terminal at a small resolution. Since Ubuntu Server has no graphical desktop, this was purely a VirtualBox window-size issue rather than an OS problem.

f. Solutions

Solution to Issue 1 . Skipping Mirror Update Step

During subsequent installations (for practice), the mirror update step was skipped in the Subiquity installer by selecting 'No updates' or by proceeding past the mirror configuration screen quickly. This significantly reduced installation time. The system can always be fully updated afterward using: sudo apt update && sudo apt upgrade

Solution to Issue 2 .Window Scaling

In the VirtualBox menu, View → Virtual Screen → Scale was used to resize the VirtualBox display window, making the terminal text larger and easier to read on high-resolution screens.

g. Filesystem Support

Ubuntu Server 24.04.4 LTS supports a wide range of filesystems, both natively and through installable kernel modules. Below is a comprehensive analysis of the major filesystem types and their relevance to Ubuntu Server:

Filesystem	Read Support	Write Support	Used by Ubuntu?	Notes
ext4	✅ Native	✅ Native	✅ Default root FS	Ubuntu's primary filesystem — journaled, stable, fast
Btrfs	✅ Native	✅ Native	⚠ Optional	Copy-on-write, snapshots, good for large servers
ZFS	✅ Via module	✅ Via module	⚠ Optional	Enterprise-grade, self-healing, requires more RAM
FAT32	✅ Native	✅ Native	⚠ EFI partition only	Required for EFI System Partition (/boot/efi)
exFAT	✅ Native (5.4+)	✅ Native	❌ Not system FS	Good for USB drives shared with Windows/Mac
NTFS	✅ Via ntfs-3g	✅ Via ntfs-3g	❌ Not system FS	Can mount Windows NTFS drives for data access
HFS+	✅ Limited	⚠ Experimental	❌ Not system FS	Apple macOS filesystem — limited Linux support
APFS	⚠ Experimental	❌ No	❌ Not system FS	Modern Apple filesystem — very limited Linux support
XFS	✅ Native	✅ Native	⚠ Optional	High-performance, good for large files

Why ext4 is the Default for Ubuntu Server

Ubuntu Server uses ext4 (Fourth Extended Filesystem) as its default root filesystem for several well-established reasons:

- Journaling: ext4 keeps a journal of filesystem changes. If the system crashes or loses power unexpectedly, the journal allows the OS to quickly recover and return the filesystem to a consistent state on the next boot — without needing a full disk scan.
- Maturity and Stability: ext4 has been the standard Linux filesystem since 2008. It has been tested across millions of deployments and is considered extremely reliable.
- Performance: ext4 uses delayed allocation and extent-based storage, which reduces fragmentation and improves write performance for both small and large files.
- Large File and Volume Support: ext4 supports individual files up to 16 TB and volumes up to 1 Exabyte, which is more than sufficient for virtually any server workload.
- Wide Tool Support: Tools like fsck, e2fsck, tune2fs, and resize2fs are mature, well-documented, and universally available on any Linux distribution.

In this installation, the disk partitioning created the following layout:

Partition	Filesystem	Mount Point	Purpose
/dev/sda1	FAT32	/boot/efi	EFI System Partition — bootloader files
/dev/sda2	ext4	/boot	Kernel and initrd images
/dev/sda3	ext4	/	Root filesystem — entire OS lives here

h. Advantages and Disadvantages

Aspect	Advantages	Disadvantages
Cost	Completely free and open-source — no licensing fees	Some enterprise features require Ubuntu Pro subscription
Security	Regular LTS security patches for 5 years; AppArmor built-in; minimal attack surface	Misconfigurations can expose services if admin lacks Linux knowledge
Performance	No graphical desktop overhead — very low memory and CPU usage (only 11% RAM in our setup)	Performance tuning requires deep knowledge of Linux kernel parameters
Software	Access to 60,000+ packages via apt; most developer tools available	Some commercial or proprietary software has no Linux version

Aspect	Advantages	Disadvantages
	natively	
Community	Enormous community, Canonical support, Ask Ubuntu, Stack Overflow	Community support quality varies; official support costs money
Compatibility	Runs on VirtualBox, VMware, KVM, Hyper-V, all major cloud platforms	Migrating between hypervisors or cloud providers can be complex
Learning Curve	Best way to develop real server admin skills — forces command-line proficiency	Steep learning curve for beginners accustomed to GUI operating systems
Stability	LTS releases are extremely stable; used by Netflix, Google, Amazon	Rolling updates in non-LTS versions can occasionally break packages

i. Conclusion

This project provided a comprehensive, hands-on introduction to installing and working with Ubuntu Server 24.04.4 LTS in a virtualized environment using Oracle VM VirtualBox. Through the process, it became clear that Ubuntu Server is not simply a desktop Linux with a missing graphical interface — it is a carefully crafted, production-grade operating system designed from the ground up for reliability, security, and efficient resource usage.

The installation itself, while straightforward in concept, required careful attention to virtualization settings (VT-x must be enabled in BIOS), disk partitioning (ext4 is used for the root filesystem because of its journaling, stability, and performance characteristics), and user account creation. The system, once running, proved remarkably lightweight: with 109 processes and only 11% memory usage, it demonstrates why server operating systems deliberately avoid graphical environments.

Working through this project also highlighted the importance of documentation — recording each step, capturing screenshots, and noting problems and their solutions. These habits are directly transferable to real-world server administration and DevOps workflows, where reproducible, documented procedures are essential for team collaboration and incident response.

j. Future Outlook / Recommendation

Looking forward, Ubuntu Server 24.04.4 LTS offers an excellent foundation to build upon in several directions:

7. LAMP/LEMP Stack Deployment: Ubuntu Server is the go-to platform for hosting web applications with Apache or Nginx, MySQL/MariaDB, and PHP. Deploying such a stack on this VM would be a natural next step.
8. Container Technologies: Docker and Kubernetes run natively and efficiently on Ubuntu Server. Exploring containerized application deployment would build on the foundation established in this project.
9. Remote Administration via SSH: Since OpenSSH was installed, the next step is configuring SSH key-based authentication (disabling password login) to practice secure remote server management from the host machine.
10. Automation with Shell Scripting: Ubuntu Server's Bash shell is a powerful automation tool. Writing shell scripts to automate backups, log monitoring, and system health checks would directly address Question 5 of this assignment.
11. Cloud Migration: The skills learned managing Ubuntu Server in VirtualBox transfer directly to managing Ubuntu Server instances on AWS EC2, Google Compute Engine, or Azure Virtual Machines — making this a directly career-relevant skill set.

As a recommendation for future students: do not skip the command-line fundamentals. Tools like grep, awk, sed, systemctl, journalctl, and cron are not optional extras — they are the primary interface through which Linux servers are managed professionally.

Virtualization in Modern Operating Systems

This section addresses the what, why, and how of virtualization in the context of modern operating systems — a technology that made this entire assignment possible.

What is Virtualization?

Virtualization is the technology that creates a simulated, software-based version of a computing resource — whether that is an entire computer, a storage device, a network, or an operating system — on top of existing physical hardware. The simulated environment is called a virtual machine (VM), and the software that creates and manages it is called a hypervisor or Virtual Machine Monitor (VMM).

In simpler terms: virtualization lets one physical computer pretend to be multiple computers simultaneously, each running its own independent operating system and software stack, completely isolated from the others. The physical machine is called the host, and the simulated machines running on it are called guests.

Key Concept: The Virtualization Layer

Physical Hardware → Hypervisor (VirtualBox) → Guest OS 1 (Ubuntu Server) + Guest OS 2 (Windows) + Guest OS 3 (Fedora) All three guest operating systems share the same CPU, RAM, and disk hardware, but each believes it has dedicated hardware. The hypervisor manages this illusion through CPU instruction interception, memory mapping, and I/O virtualization.

Why Virtualization? (The Motivations)

Modern organizations and individuals adopt virtualization for the following reasons:

1. Hardware Consolidation and Cost Reduction

Before virtualization, a company might need ten separate physical servers for ten different applications. Each server ran at 10–20% of its capacity most of the time, wasting electricity, cooling, and physical space. With virtualization, all ten workloads

can run on two or three powerful physical servers at 70–80% capacity, dramatically reducing hardware purchasing, power consumption, and data center space costs.

2. Isolation and Security

Each virtual machine is completely isolated from every other VM on the same host. A virus, crash, or security breach in one VM cannot directly affect another VM or the host. This isolation makes VMs ideal for running untrusted code, testing potentially dangerous software (such as malware analysis), and providing separate environments for different users or departments.

3. Snapshots and Rollback

Hypervisors like VirtualBox allow the administrator to take a snapshot — a frozen copy of the entire VM state at a specific moment. If an update breaks the system, an experiment goes wrong, or ransomware encrypts the disk, the administrator can instantly roll back to the last good snapshot. This capability is impossible with physical machines without expensive backup hardware.

4. Portability and Migration

A virtual machine is essentially a set of files on the host's disk. This means an entire server environment — OS, applications, configurations, and data — can be copied, zipped, and moved to a different physical machine in minutes. This portability enables fast disaster recovery and facilitates migration between data centers or cloud providers.

5. Development and Testing Environments

Developers can create isolated VMs that exactly replicate a production server environment. This eliminates the classic 'it works on my machine' problem. Teams can also test software on multiple operating systems simultaneously without needing multiple physical devices — exactly as demonstrated in this assignment.

6. Cloud Computing Foundation

The entire cloud computing industry is built on virtualization. When you rent a virtual machine from AWS (EC2), Google Cloud (GCE), or Azure (VM), you are renting a virtual machine running on Xen, KVM, or Hyper-V hypervisors in a massive data center. Without virtualization, cloud computing as we know it would not exist.

How Virtualization Works

Virtualization operates through several interconnected technical mechanisms:

The Hypervisor

The hypervisor is the core software of virtualization. There are two types:

	Name	Description	Examples
Type 1 (Bare-Metal)	Native Hypervisor	Runs directly on the physical hardware, with no host OS between it and the hardware. Most efficient and used in enterprise and cloud.	VMware ESXi, Microsoft Hyper-V, Xen, KVM
Type 2 (Hosted)	Hosted Hypervisor	Runs as an application on top of a host OS (e.g., Windows or macOS). Less efficient but easier to use. Used for development and learning.	Oracle VM VirtualBox, VMware Workstation

Oracle VM VirtualBox — used in this assignment — is a Type 2 hypervisor. It runs on top of Windows (the host OS) and creates virtual machines, including the Ubuntu Server VM documented in Question 1.

CPU Virtualization

Modern CPUs from Intel (VT-x) and AMD (AMD-V) include hardware support for virtualization at the processor level. When the hypervisor runs a guest VM, the guest OS's privileged CPU instructions (which normally require direct hardware access) are intercepted by the hypervisor and safely executed in a controlled sandbox. Without hardware virtualization support (VT-x/AMD-V), software-only virtualization is much slower — which is why enabling VT-x in BIOS was necessary in this project.

Memory Virtualization

Each virtual machine gets its own virtual address space. The hypervisor uses techniques like Extended Page Tables (EPT) on Intel or Rapid Virtualization Indexing (RVI) on AMD to maintain a mapping between the guest VM's virtual memory addresses and the actual physical RAM addresses on the host. This allows multiple VMs to each believe they own contiguous blocks of memory, even though they are actually sharing the same physical RAM chips.

Storage Virtualization

Virtual machines use virtual hard disk files instead of real physical disks. In VirtualBox, the virtual disk is stored as a .vdi (VirtualBox Disk Image) file on the host's filesystem. The VM sees it as a real physical disk (SATA or NVMe). The hypervisor translates disk read/write operations from the guest OS into file operations on the host filesystem. This is why the Ubuntu Server installation created partitions (/, /boot, /boot/efi) inside what is actually a single file on the Windows host.

Network Virtualization

VirtualBox provides several networking modes for VMs:

- NAT (Network Address Translation): The VM shares the host's IP address. The VM can access the internet, but external machines cannot directly reach the VM. Used in this assignment — hence IP 10.0.2.15.
- Bridged: The VM gets its own IP address on the local network, appearing as a separate physical device. Useful when other devices need to connect to the server.
- Host-Only: Creates a private network between the host and VMs only. No internet access. Useful for isolated lab environments.
- Internal: VMs can only communicate with each other — no host or internet access. Used for simulating isolated multi-server networks.

Paravirtualization and Hardware Passthrough

Modern hypervisors also support paravirtualization — where the guest OS is modified to be 'aware' that it is running in a VM and cooperates with the hypervisor for better performance. Linux VMs in VirtualBox use paravirtualization interfaces (like KVM paravirtualization) for better clock accuracy and I/O performance. For tasks requiring direct hardware access (GPU passthrough, USB devices), VirtualBox can pass through specific physical hardware devices directly to a VM.

Summary: Virtualization in Three Lines

WHAT: A technology that creates isolated, software-defined computers (VMs) on top of physical hardware. WHY: Cost reduction, isolation, snapshots, portability, development environments, and cloud computing. HOW: Through a hypervisor that intercepts and manages CPU, memory, storage, and network resources, distributing them safely among multiple guest operating systems.

REFERENCES

- Canonical Ltd. (2024). *Ubuntu Server Documentation*. Available at: https://ubuntu.com/server/docs
- Ubuntu. (2024). *Download Ubuntu Server 24.04 LTS*. Available at: https://ubuntu.com/download/server
- Oracle. (2024). *VirtualBox User Manual*. Available at: https://www.virtualbox.org/manual
- Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). *Operating System Concepts* (10th ed.). Wiley.
- Tanenbaum, A. S., & Bos, H. (2015). *Modern Operating Systems* (4th ed.). Pearson.
- AWS Documentation. (2024). *Introduction to Virtual Machines*. Available at: https://docs.aws.amazon.com